changes its state in response to the agent's actions. Environments can be simulated (like a game) or real-world (like a robotic arm).

3. **Reward Signal**:
    - A scalar value that guides the agent’s learning process. Positive rewards encourage desirable behavior, while negative rewards discourage undesirable actions. For instance, a robot successfully reaching its target might receive a positive reward, while colliding with obstacles results in a penalty.

The overarching goal of the agent is to learn an **optimal policy** that maximizes the cumulative reward over time. Achieving this requires balancing exploration (trying new actions to discover rewards) and exploitation (choosing actions known to yield high rewards).

Essential Concepts in Reinforcement Learning

1. **State**:
    - Represents the current situation or configuration of the environment. It provides the agent with all the necessary information to make decisions. For example, in a chess game, the state includes the positions of all the pieces on the board.
2. **Action**:
    - The set of possible moves the agent can take in a given state. For example, a robot might move forward, backward, or rotate.
3. **Policy (π)**:
    - A strategy or mapping that defines the agent’s actions based on the current state. Policies can be deterministic (always selecting the same action for a state) or stochastic (selecting actions probabilistically).
4. **Value Function (V)**: